

MAT 3003 001

Discrete Mathematics

Course Syllabus - Spring 2026



Course Information

Course Description: Prerequisite: MAT 1313 or CS 2233 or consent of instructor. The course may include (1) Propositional logic, (2) Predicate Logic, (3) Sets and Boolean algebras, (4) Relations, ordered sets, and bounds, (5) Functions, operations of functions, and their images and inverses, (6) Well-ordered sets, induction and choice functions, and (7) Introduction to computability, classical and contemporary models of computation, and limitations of computation. This course has Differential Tuition.

Credit Hours: 3

Course Modality: Traditional in-person

Meeting Times

Duration: 01/20/2026-05/13/2026

Campus: Main Campus

Location: [MS 2.02.54](#)

Time(s): MWF 2:00 PM - 2:50 PM

Learning Goals

At the end of the course, you will be able to:

Relations

Outcome

Understand basic properties of relations, both discrete and real-valued

Advanced combinatorics and large networks

Outcome

Understand the importance of graph colorings and Ramsey theory (finitary, infinitary and real-valued)

Alphabets and languages

Outcome

Identify quickly when a given set is countable or uncountable based on its description

Ordinal numbers

Outcome

Understand ordinal arithmetic and transfinite induction

Cardinal numbers

Outcome

Understand cardinal addition, multiplication, and exponentiation

Limits of structures

Outcome

Understand the applications of graph limits, transfer principles, and Los' theorem

Introduction to logic

Outcome

First-order logic and compactness. Connections with combinatorics and Ramsey theory

Skills Goals

At the end of the course, you will be able to:

At the end of the course, you should be comfortable understanding the interface between continuous and discrete mathematics, and the interface between infinite and finite sets via limits and transfer principles.

You should be comfortable with measures of infinity (ordinal and cardinal numbers) and the foundations of mathematics (the Continuum Hypothesis and the Axiom of Choice)

Communicate with Me

Instructor Name: Jose Iovino

Department

Mathematics

Office Location

FLN 4.01.13

Student Hours

Mondays and Wednesdays, 4:00-6:30 pm

Email Address: jose.iovino@utsa.edu

CV Link

<https://iovino.us/>

Preferred Method of Communication

Email

About Me & My Teaching Philosophy

I will assign homework exercises after each lecture. It is important that you work on the homework problems the same day they are assigned. Do not postpone your work. Have the exercises ready to turn in the day after they are assigned. The course moves fast, and if you do so, you would risk getting completely lost.

You should have at least two hours per day to work on the exercises alone. (This doesn't include the time that you have to spend studying the material in the textbook.)

Policy on take-home projects: The solutions that you submit to the assigned problems must be your own, and you must be ready to answer questions about them. If you are unable to answer questions about your approach to a particular problem, you cannot get credit for that problem.

It is a violation of the Student Code of Conduct to present the ideas or work of others as your own work.

Course Materials

A First Course in Contemporary Discrete Mathematics

Additional Information
Available on CoCalc

To enhance your learning and adequately prepare for all assignments and assessments, please consider all course materials listed above, along with additional content provided in class, posted to our course site, or included in the Course Schedule, as required reading and study unless indicated as optional.

Assessments and Assignments

I will assign homework exercises after each lecture. It is important that you work on the homework problems the same day they are assigned. Do not postpone your work. The course moves fast, and if you do so, you would risk getting completely lost.

You should have at least two hours per day to work on the exercises alone. (This doesn't include the time that you have to spend studying the material in the textbook.)

Policy on take-home projects: The solutions that you submit to the assigned problems must be your own, and you must be ready to answer questions about them. It is a violation of the Student Code of Conduct to present the ideas or work of others as your own work.

Important: You should be prepared to answer questions about your proofs at any time. If you are unable to answer questions about a logical argument used in your exam satisfactorily, you will be penalized for the exam and you not receive a passing grade.

Policy on all written work: Your work should be shown completely, using correct notation, in such a way that any person with basic knowledge of the material can follow your argument fully, without confusion. Solutions must be typeset in TeX or LaTeX. Disorganized work, or work not typeset in LaTeX will not be graded.

Policy on work for graduate credit: The course is cross-listed in both the graduate and undergraduate catalog. Students taking the course for graduate credit will be required to do extra work. In particular, these students will be required to prepare the lecture notes, professionally presented in LaTeX, and to make them available for the rest of the class. The presentation of the lecture notes should be of professional standards. I will meet regularly with the graduate students to ensure the highest quality.

Activities and Grading

Activity	Quantity	%
Take home exams	4	80
Quizzes	Ideally, one per week	10
Presentations and participation		10
Total		100%

Distribution of Course Assignments, Their Quantity, and Contribution to Final Grade.

Grade Distribution and Letter Grade

Letter Grade	Grade Range
A	[90,100]
B	[80,90)
C	[70,80)
D	[60,70)
F	<60

Course Guidelines for the Use of Generative Artificial Intelligence

I do not recommend use of large language models for the projects. AI tools, although extremely useful for other tasks, are still too primitive for this type of mathematics. If you rely heavily on them, they will make you look like a fool. Using AI to do abstract mathematics would be like using a Phillips head screwdriver for a flathead screw.

Important: You should be prepared to answer questions about your proofs at any time. If you are unable to answer questions about a logical argument used in your exam satisfactorily, you will be penalized for the exam and you not receive a passing grade.

Course Expectations & Policies

Instructor-Initiated Drops

This course uses instructor-initiated drops for students who exceed the absence and/or missed assignment limit. Therefore, up to the last day for students to withdraw from an individual course, October 27, you will be dropped for exceeding **6 absences or one missing assignment**. Students will receive at least one courtesy warning when approaching the absence/missed assignment limit. Notification will be sent using ASAP to the student's email address. A subsequent absence or missed assignment will result in being dropped from the course. Notification of being dropped will also be sent through ASAP to the student's email address. *This drop does not affect enrollment in other courses.* Please consult the [Dropping Courses webpage](#) for further details on the process and appeals.

Course Schedule

For a list of important university-wide dates, review [One Stop's academic calendar](#).

Department Policies & Resources

Ombudsperson

The ombudsperson is an advocate who investigates complaints about a specific section or instructor and attempts to resolve them through mediation. Comments or complaints sent to this person will be brought to the attention of department leadership, who will follow up. If you have questions about the logistics of your class, please contact the instructor.

Contact Ilse Rosales Martinez at ilse.rosalesmartinez@utsa.edu

Tutoring Services

UTSA provides many opportunities for student success, including tutoring services for specific math courses. The list below is subject to change based on availability.

MAT 1023, MAT 1043, MAT 1053, and MAT 1073

- We offer peer-to-peer academic support to students enrolled in a variety of courses. Both drop-in tutoring and one-on-one appointments are available.
- **Location:** [Tomás Rivera Center for Academic Excellence](#) MS 2.02.18 (behind the Rowdy statue)

MAT 1053 and MAT 1073

- Synergistic Targeted Academic Review Sessions (STARS), peer-led, interactive, and group focused sessions which cover traditionally difficult topics. The schedule will be provided by instructors through Canvas.
- **Location:** Determined by STARS tutors

MAT 1093, MAT 1213/1214, and MAT 1223/1224

- The Math Gym is unique in that it's a WeBWoRK lab designed to offer our STEM core students support from math majors and minors through targeted SI sessions. The schedule will be provided by instructors through Canvas
- **Location:** [Math Gym FLN 03.02.08](#)

Note: Course-specific support for upper-division courses may be available for a particular semester. Please request this information from your instructor.

Essential Student Information

- **Important:** Bookmark and visit the [Common Syllabus Information webpage](#) to find important and valuable resources about counseling services, transitory/minor medical issues, supplemental instruction, tutoring services, academic success coaching, sexual harassment and sexual misconduct, campus safety and emergency preparedness, and the Roadrunner Creed.
- For technical requirements, support, and resources, visit [Academic Innovation's Student Technical Support](#) page.
- UT San Antonio provides reasonable accommodations to students via [Student Disability Services](#). For more details on eligibility, policies, and requirements, please visit [Applying for Accommodations](#) or call (210) 458-4157.
- Your well-being is important. In addition to academic growth, life as a student can include many emotional experiences and opportunities. Being mindful of your own needs and available support can provide you with the tools to successfully navigate these experiences. Student support for well-being, including [24/7 mental health services](#), can be accessed from the [Well-being at UT San Antonio](#) site for students. If needed, we can schedule a meeting, and I can connect you to resources. You are not alone; I am available to help you find resources that might best meet your needs.
- [Student Assistance Services \(SAS\)](#) is available to help you navigate academic and personal challenges. Campus Advocates in SAS are a central point of contact for students seeking clarity on university processes and policies. They also provide a direct connection to helpful resources and advocacy rooted in care. Campus Advocates offer confidential support and personalized guidance to help students succeed.

- The [Roadrunner Pantry](#) is here to help UT San Antonio students stay healthy and focused by providing free access to groceries, toiletries, and other essentials. With locations at the Main and Downtown campuses, the pantry is organized like a small store and offers items such as canned goods, snacks, and hygiene products. Students with a valid UTSA ID can visit the pantry to pick up what they need. Our goal is to ensure every Roadrunner feels supported and has the resources to succeed. If you're ever short on food or supplies, the Roadrunner Pantry is a welcoming space you can count on.
- Students at UT San Antonio are responsible for ensuring their work is consistent with UT San Antonio's standards for academic integrity. Students should review [Section 203 of the UT San Antonio Student Code of Conduct](#) for appropriate standards of academic integrity.
- UT San Antonio provides numerous services for students from counseling to tutoring to a food pantry. Visit [Student Affairs Programs and Services](#) and [Student Success](#) for more information.
- Visit the [UT San Antonio Libraries and Museums](#) site for access to journals, research tutorials, and tech gear you can borrow and to find your department's librarian.
- Enroll in the [Roadrunner Success Playbook](#), an open-enrollment, self-paced, online hub in Canvas tailored to ensure you have the resources you need to excel at UT San Antonio.
- Follow [Digital Learning Netiquette](#) standards for your online communication activities.